Description of the Fair H4 Subspace Benchmark Experiment

P765 Final Project

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1 Purpose

This experiment compares three different basis-state families for quantum subspace methods on the H_4 molecule:

- Krylov basis states,
- UCCSD basis states,
- fixed-architecture hardware-efficient ansatz (HEA) basis states.

The goal is to compare these three families fairly as *subspace bases*. Each family generates a small set of states, and those states define a projected Hamiltonian problem whose lowest eigenvalue approximates the ground-state energy.

2 Hamiltonian and Reference State

The H_4 benchmark uses the PennyLane quantum chemistry Hamiltonian for the default linear-chain H_4 geometry already used in the project:

- molecule: H_4,
- bond length: 0.74 Å,
- basis: `sto-3g`,
- unit: `angstrom`.

For this system, the notebook builds:

- the qubit Hamiltonian $\hat{H}$,
- the Hartree–Fock reference state,
- the electron number and qubit count,
- the exact FCI reference energy in the fixed-electron sector.

In the validated default run, H_4 maps to 8 qubits, 4 electrons, and a Hamiltonian with 185 Pauli terms.

3 Basis Families

Krylov basis

Each Krylov state is generated from the Hartree–Fock reference using Trotterized time evolution:

$$|\phi(t)\rangle \approx e^{-it\hat{H}}|\phi_{\mathrm{HF}}\rangle.$$

One scalar time parameter is optimized for each added basis state.

UCCSD basis

Each UCCSD state starts from the same Hartree–Fock reference and uses PennyLane's `qml.UCCSD` template. For H_4, the default setup produces 26 trainable UCCSD parameters for each optimized basis state.

Fixed-architecture HEA basis

Each HEA state also starts from the Hartree–Fock reference. The circuit architecture is fixed globally:

- one layer,
- rotation gates `RY` and `RZ`,
- `CNOT` entanglers,
- chain connectivity.

Only the angle vector is optimized. For H_4, this gives 16 trainable HEA parameters per optimized basis state.

4 Common Subspace Objective

If the benchmark constructs basis states

$$\{|\phi_0\rangle, |\phi_1\rangle, \ldots, |\phi_{K-1}\rangle\},$$

then all families are compared using the same projected matrices:

$$H_{ij} = \langle\phi_i|\hat{H}|\phi_j\rangle, \qquad S_{ij} = \langle\phi_i|\phi_j\rangle.$$

The projected ground-state energy is the lowest generalized eigenvalue from

$$H\mathbf{c} = E_{\mathrm{sub}} S\mathbf{c}.$$

The same overlap-conditioning rule is used for all three families. In the current H_4 package, a looser overlap cutoff is used so that nearly dependent states can still be retained when the generalized eigensolver remains numerically manageable. If the overlap matrix develops eigenvalues below the cutoff

$$10^{-10},$$

the candidate basis state is rejected and receives a large penalty. This prevents nearly duplicate states from unfairly enlarging the basis.

5 Fairness Protocol

The H_4 benchmark follows the same fairness rules as the original H_2 notebook:

- same Hamiltonian,
- same Hartree–Fock reference state,
- same basis size target K,
- same projected-subspace objective,
- same optimizer family and budget,
- same overlap-conditioning rule.

The basis is built greedily:

1. the first basis state is the Hartree–Fock state,
2. one new basis state is optimized at a time,
3. previously selected basis states remain frozen,
4. the projected energy is recomputed after each addition.

6 Default H_4 Notebook Configuration

The molecule-specific notebook `Fair_Subspace_Benchmark_H4.ipynb` uses conservative defaults so the benchmark remains runnable:

- `mol_name = "H4"`,
- `bond_length = 0.74`,
- `basis = "sto-3g"`,
- `unit = "angstrom"`,
- `K_max = 6`,
- `device_name = "lightning.qubit"`,
- `optimizer_method = "Powell"`,
- `maxiter = 20`,
- `maxfev = 120`,
- `n_restarts = 2`.

The remaining fairness-related settings match the H_2 benchmark:

- `random_seed = 1234`,
- `t_max = 1.0`,
- `theta_max = 1.0`,
- `s_eval_cutoff = 1e-10`,
- one-layer fixed HEA with `RY`/`RZ` rotations and chain `CNOT` entanglers.

7 Default H_4 Results

For the validated default run, the FCI reference energy is

$$E_{\mathrm{FCI}} = -2.138889915372 \text{ Ha}.$$

The projected subspace energies across the default sweep are:

Family	K	E_{sub} (Ha)	$E_{\mathrm{sub}} - E_{\mathrm{FCI}}$ (Ha)
Krylov	1	-2.097899614513	$+0.040990300858$
Krylov	2	-2.132228489342	$+0.006661426030$
Krylov	3	-2.134646254913	$+0.004243660459$
Krylov	stopped before $K = 4$ due to loss of a valid new basis state		
UCCSD	1	-2.097899614513	$+0.040990300858$
UCCSD	2	-2.101496822492	$+0.037393092880$
UCCSD	3	-2.106269604486	$+0.032620310886$
UCCSD	4	-2.114618877346	$+0.024271038026$
UCCSD	5	-2.120300338326	$+0.018589577046$
UCCSD	6	-2.126235550356	$+0.012654365016$
Fixed HEA	1	-2.097899614513	$+0.040990300858$
Fixed HEA	2	-2.098043767103	$+0.040846148269$
Fixed HEA	3	-2.098651296353	$+0.040238619019$
Fixed HEA	4	-2.099119552287	$+0.039770363085$
Fixed HEA	5	-2.100326701779	$+0.038563213593$
Fixed HEA	6	-2.100708614230	$+0.038181301142$

Under the same fair subspace protocol:

- Krylov gives the strongest early improvement and now extends to $K = 3$ under the looser cutoff, but it still saturates before $K = 4$.
- UCCSD continues to improve as K increases from 2 to 6, giving the best final energy among the families that complete the full sweep.
- the fixed HEA also improves monotonically through $K = 6$, but much more slowly under the same budget.

8 Interpretation

This H_4 experiment answers the question:

> Given the same H_4 Hamiltonian, the same Hartree–Fock starting point, the same subspace size, and the same optimization budget, which basis family gives the best projected ground-state energy?

For the conservative default settings used here, the H_4 gap-vs-K story is now clearer. The looser cutoff allows Krylov to add one more nearly dependent state and reduce the gap further by $K = 3$, but Krylov still saturates early. UCCSD keeps improving steadily all the way through $K = 6$, while the fixed HEA also improves but only modestly. In this larger sweep, UCCSD is the strongest family for continued basis growth, whereas Krylov remains the strongest low-K option.

9 Implementation Files

The H_4 benchmark package is implemented in:

- `Fair_Subspace_Benchmark_H4.ipynb`
- `Fair_Subspace_Benchmark_H4_Description.tex`